

Muhammad Ibrahim

(+92) 309-424-2532 · ibrahimanwarulhaq1994@gmail.com · LinkedIn · Portfolio

TECHNICAL SKILLS

- **AI & LLM Systems:** GPT-4, LLM Integration, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Function Calling, Conversation Memory, OpenRouter
- **Backend Development:** FastAPI, Node.js, RESTful APIs, Redis, WebSockets, JWT Authentication, Docker
- **Frontend Engineering:** React.js, Next.js, TypeScript, Responsive Web Design
- **Languages:** Python, JavaScript, SQL (PostgreSQL, MySQL)
- **Data Systems & Tools:** ETL Pipelines, Data Validation, Git, Tableau, SSIS

EXPERIENCE

IBEX

Software Engineer

Lahore, Pakistan
Nov 2024 – Present

- Improved the ID&V flow for an enterprise client's AI customer service agent on Sierra's platform by implementing a multi-step fallback system – verifying users by plan number and email when phone verification failed – increasing overall verification success rate.
- Engineered full-stack features across React.js and FastAPI as 1 of 5 engineers on AMP, IBEX's internally built AI Agent Builder Platform, enabling internal teams to configure, test, and deploy production-grade voice and chat agents.
- Designed and implemented Redis-based session management and conversation history tracking on AMP, powering context-aware LLM interactions for agents handling **2,000+ calls/month** at an **84% containment rate**.
- Collaborated with AI and data teams to integrate RAG-based knowledge pipelines into AMP's agent infrastructure, enabling domain-aware responses across enterprise deployments.

PROJECTS

AI Agent Builder Platform (AMP)

Platform Link

Tech Stack: Python, FastAPI, React.js, OpenAI API, Redis, WebSockets, Docker

- Built full-stack features for AMP, a production AI agent platform supporting voice and chat interfaces, used by internal prompt engineers to deploy agents handling **2,000+ monthly interactions** at **84% containment**.
- Developed conversational workflow features integrating LLM function calling and RAG knowledge pipelines, enabling domain-aware agent responses across enterprise use cases.
- Containerized services using Docker to ensure consistent, reproducible environments across development and production deployments.

MeetBot – AI-Driven Meeting Assistant

GitHub · Live Demo

Tech Stack: Next.js, TypeScript, FastAPI, RAG, OpenRouter API, Notion API

- Built an end-to-end AI meeting assistant that ingests raw transcripts and converts them into a structured, searchable knowledge base using RAG and LLM summarization.
- Implemented overlapping character-window chunking with configurable chunk size and overlap to preserve context continuity across transcript boundaries, enabling coherent retrieval over long meeting transcripts.
- Implemented vectorized retrieval with LLM-driven summarization via OpenRouter API and engineered asynchronous background processing jobs for high-throughput transcript ingestion.
- Integrated Notion API to automatically export structured meeting summaries and action items post-processing, reducing manual follow-up overhead.

EDUCATION

National University of Sciences and Technology (NUST)

Bachelor of Science in Computer Science (CGPA: 3.42 / 4.0)

Islamabad, Pakistan
Oct 2020 – Jun 2024

- **Final Year Project:** Suspicious Behavior Detection using CCTV – Awarded **3rd Position** at the Department Industrial Open House for excellence in real-time computer vision modeling.